



On the predictability of ETF returns with technical predictors

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ABSTRACT

We develop a machine learning framework that uses technical indicators derived from international stock data to forecast the performance of international equity ETFs. To address the limited history of ETFs, we train a random forest model on global stock data and apply it to rank ETFs according to the probability of outperformance. Portfolios formed on this ranking are associated with economically meaningful gross-of-cost return spreads, averaging 0.76% per month (t -stat = 2.76) and 0.62% after market risk adjustment. Predictability is strongest at short horizons and decays with longer holding periods. Volatility and momentum indicators contribute the most to model performance. In line with limits to arbitrage, predictive strength is more pronounced in less efficient markets and among lower-liquidity ETFs. Results are robust across portfolio construction methods and alternative models, and the signal performs well out-of-sample from 2011 to 2022, with extended evidence up to 2024.

1. Introduction

Studying the cross-sectional predictability of exchange-traded funds (ETFs) returns is interesting for individual investors, portfolio managers, and policymakers, given the growing importance of ETFs in the global financial markets. ETF assets under management (AUM) rose nearly three times faster than mutual funds between 2016 and 2022.¹ ETFs now span a wide array of asset classes, including equities, bonds, and commodities. In this study, we focus specifically on equity ETFs, which are the most liquid and widely traded segment of ETFs. However, their relatively short return histories limit the development of robust out-of-sample prediction models.

To overcome this data constraint, we explore whether information from individual stocks, whose return histories span several decades, can be used to predict cross-sectional return differences. Since equity ETFs represent tradable baskets of individual stocks, their cross-sectional performance is mechanically linked to the cross-sectional dispersion of the underlying securities. However, whether predictive signals extracted from stock-level return dynamics can transfer to ETF-level cross-sectional rankings is not trivial. Such transfer would require persistence in relative performance at the basket level and incomplete arbitrage of stock-level signals in ETF prices. As a result, the cross-sectional behavior of individual securities may embed predictive signals that carry over to the ETFs they comprise. We build a framework in which models are trained on stock-level technical signals and tested on equity ETFs, can be viewed as tradable wrappers around diversified stock baskets. This cross-asset approach allows us to evaluate whether predictive signals extracted from individual securities can be transferred to basket-level returns when the basket itself has a limited return history. Technical indicators, such as moving averages and momentum measures, are especially attractive because they are simple, broadly applicable across assets, and potentially effective in capturing short-term return patterns. They are widely used by

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¹ <https://www.oliverwyman.com/our-expertise/insights/2023/may/exchange-traded-funds-are-fueling-market-opportunities.html>.

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practitioners (e.g., [Menkhoff, 2010](#)) and remain relevant in academic forecasting studies (e.g., [Lo et al., 2000](#); [Nti et al., 2020](#)). We hypothesize that technical indicators derived from individual stocks may be reused to forecast ETF performance.

Our dataset includes individual stock data from 1981 to 2021 and equity ETF data from 2005 to 2024, all sourced from Datastream. It covers over 9000 ETFs from 52 countries. To test our hypothesis, we build tree-based machine learning models, applying technical indicators derived from stock data to forecast ETF returns. Our analysis has two primary goals: first, to assess whether technical signals derived from stock data can effectively forecast ETF performance; and second, to examine how this predictability varies across countries and the practical constraints of such strategies by analyzing portfolio turnover and ETF liquidity.

We employ random forests, a tree-based machine learning model, to predict international ETF performance. This method is well-suited to our task due to its robustness, computational efficiency, and effectiveness in handling noisy financial data. It also provides built-in feature importance measures that help us interpret which technical signals drive the predictions. Our choice is supported by previous research that demonstrates the superior performance of tree-based models in return prediction tasks (e.g., [Ballings et al., 2015](#); [Gu et al., 2020](#); [Leippold et al., 2022](#); [Avramov et al., 2023](#)). Rather than predicting return magnitudes, we predict cross-sectional outperformance probabilities. Our objective is not classification accuracy per se, but the generation of economically meaningful cross-sectional rankings that produce return spreads in portfolio sorts. Thus, we adopt the classification-based approach following [Breitung \(2023\)](#), which labels ETFs as outperformers if their returns exceed the cross-sectional median in each month. Prior work (e.g., [Basak et al., 2019](#); [Nabipour et al., 2020](#)) shows that binary classification can improve model performance relative to regression, particularly in financial contexts. Using the models-implied rankings as investment signals, we construct long–short portfolios by taking equal-weighted long positions at the top 10% and short positions in the bottom 10%.

Our results show that technical indicators constructed from stock-level data contain substantial cross-sectional predictive information for future ETF outperformance probabilities. Specifically, models trained on stock-based technical indicators can predict ETF outperformance probabilities, generating gross-of-cost (i.e., before transaction costs and other trading frictions) long–short return spreads of 0.76% (t -stat = 2.76) in local currency, and after adjusting for market risk factor, the mean return is 0.62% with a t -statistic of 2.27. Currency effects do not drive the results; the long–short return in USD is 0.73% with a t -statistic of 2.78. In line with previous research that highlights the short-term efficacy of technical signals, we observe that such predictability is most pronounced in the early months as we increase the rebalance interval from one month to 36 months. Moreover, we analyze feature importance and then re-train models using individual categories of technical indicators. None of the single-category models match the performance of the full model, suggesting that different indicator types offer complementary predictive signals.

We also show that the greater the efficiency of a market traded by the underlying ETFs, the more challenging it becomes to predict future ETF performance. The associated gross-of-cost return spreads are larger for ETFs investing in China (0.87%) than in the United States (0.48%). To assess the implementability of these strategies, we further examine turnover levels. Even when portfolio turnover is reduced through partial rebalancing, the long–short strategy continues to generate strong and statistically significant gross-of-cost return spreads. When we apply the sticky strategy using 40% as the retention ratio, the long–short mean return is 0.58% with t -statistic 2.34. Moreover, liquidity emerges as a key factor in return predictability. Specifically, we find that long–short portfolio returns are substantially higher for low-liquidity ETFs, consistent with the notion that such markets are less efficient due to higher arbitrage costs, greater transaction frictions, and potentially slower information diffusion. At the same time, these frictions imply higher implementation costs and capacity constraints, so we interpret the reported spreads as gross-of-cost evidence of pricing frictions rather than as frictionless arbitrage opportunities.

Our paper contributes to several literature streams. Despite the growing literature on the predictability of ETFs, key gaps remain regarding the transferability of signals across assets, signal types, cross-country variation, and the robustness of forecasting models. To study ETF return predictability, a wide range of models, from traditional linear regression (e.g., [Rompotis, 2010, 2011](#); [Brown et al., 2021](#)) to the latest machine learning and deep learning techniques (e.g., [Versace et al., 2004](#); [Liew and Mayster, 2017](#); [Day and Lin, 2019](#)), have been suggested. [Rompotis \(2011\)](#) runs dummy regression analysis to show that the performance of ETFs can be somewhat predicted. Alternatively, [Chen and Kuo \(2006\)](#) propose a neural network-based decision support system to provide investors with suggestions on transaction timing and transaction strategies of Taiwan 50 index ETFs. [Zhong and Enke \(2019\)](#) predict the daily return direction of the SPDR S&P 500 ETF using a deliberately designed classification mining procedure based on hybrid machine learning algorithms. Machine learning techniques, as the latest advancements, are capable of dealing with complexity and non-linearity in data, and are especially suitable for large datasets. Yet, to our knowledge, no other study applies stock-based technical indicators to predict international ETF returns using machine learning.

Meanwhile, predictors are commonly grouped into technical indicators, which are derived from historical prices and trading patterns, and non-technical indicators, which may capture demand flows and investor sentiment. [Chen and Kuo \(2006\)](#) employ various technical indicators to build ETF transaction strategies in Taiwan. Including three US ETFs and six emerging market ETFs, [Hsu et al. \(2010\)](#) examine the predictive ability of technical trading rules: moving averages (MA) rules and filter rules (FR) and find that technical analysts in large institutions may be able to make profits over risk premiums in the ETF market. On the other hand, non-technical indicators for ETFs refer to factors not rooted in historical price and volume data, for example, fund flow and market sentiment. [Brown et al. \(2021\)](#) find that non-fundamental demand shocks, identified through ETF flows, can predict future ETF returns. [Lee et al. \(2021\)](#) examine the impact of investor attention proxied by Google Search Volume Index and home country-specific factors on ETF returns. However, there is no clear evidence of whether the technical indicators are superior to the non-technical predictors or vice versa.

While extensive literature explores ETF return predictability, most studies focus narrowly on individual markets or a limited set of ETFs. For instance, [Yang et al. \(2010\)](#), [Rompotis \(2011\)](#), [Zhong and Enke \(2019\)](#), [Brown et al. \(2021\)](#) and [Chen \(2023\)](#)

concentrate on U.S. ETF return predictability. Jares and Lavin (2004) investigate foreign ETFs (Japan and Hong Kong) that trade on U.S. exchanges but provide broad exposure to foreign markets. Day and Lin (2019) develop a robo-advisor with different machine learning and deep learning forecasting methodologies in the Taiwan ETF market. There is relatively little research on international ETFs. Broman (2020) investigates the return predictability for 4560 twin-pairs of Exchange-Traded Funds (ETFs) from 15 country pairs. Zhang et al. (2023) use 29 country-/region-specific ETFs (including US). With the new predictor, risk-neutral moments of returns, they predict future excess returns of country-/region-specific ETFs. Lee and Chen (2020) use 132 country-specific ETFs from 45 countries and examines whether social media (Twitter) happiness sentiment and country-level happiness sentiment indices predict cross-border ETF returns. Our empirical study compiles a dataset of 9,255 equity ETFs from 52 countries, which, to our knowledge, represents one of the most comprehensive cross-country equity ETF datasets assembled to date.

Our study makes two key contributions to the literature on return predictability and cross-asset signal transfer. First, we analyze international equity ETFs with the random forest classification model. Unlike previous research that often focuses on a single market (e.g., US ETF market, Chinese ETF market) or specific ETFs (e.g., energy ETFs, gold ETFs), our analysis has a broader global perspective. This provides a generalizable understanding of ETF return predictability and reveals cross-country differences in signal effectiveness. Second, we introduce an innovative approach by training our machine learning model on stock market data and subsequently applying it to ETFs, thereby addressing the challenge posed by the short historical data availability for ETFs. By using this approach, we demonstrate the potential of utilizing historical stock market data to improve the measurement of cross-sectional predictability in ETF returns. Furthermore, our study diverges from conventional return prediction approaches by focusing on the relative ranking of ETFs. This ranking-based approach offers a clearer view of cross-sectional return potential. This emphasis on predicting outperformance probabilities, rather than absolute values, enhances the precision of our assessment of cross-sectional predictability and is less time-consuming when a large dataset is applied.

Throughout, we treat ETFs primarily as transparent and scalable implementation vehicles for basket-level returns, rather than as economically distinct return-generating assets. While we evaluate portfolio return spreads to illustrate the economic strength of the predictive signals, our analysis is conducted in gross-of-cost terms and does not model detailed execution frictions or market impact. Therefore, we interpret our findings as evidence of statistical cross-sectional return predictability and cross-asset signal transfer rather than as claims of fully scalable net-of-cost trading profits.

The remainder of the paper is structured as follows: Section 2 describes the data, construction of technical indicators, machine learning methodology, and portfolio construction. Section 3 presents the main empirical results, including return predictability, feature importance, geographic variations, and implementation considerations. Section 4 offers a series of robustness checks, covering alternative models, bond ETFs, the latest period, and alternative portfolio construction methods. Finally, Section 5 concludes.

2. Data and methodology

2.1. Data sources and preprocessing

We collect international daily individual stock and equity ETF data across different exchanges from Datastream, including variables such as open price, high price, low price, closing price, turnover by volume, and total return index. For our baseline analysis, all prices and returns are denominated in local currency and are unadjusted for share splits. We select monthly technical indicators from daily data, and we calculate gross returns as monthly percentage changes in the total return index (Datastream code: RI). The final stock dataset comprises monthly observations of individual stock returns from February 1981 to June 2021, and the ETF dataset covers the period from January 2005 to December 2024.² We collect ETF data from 2005 onwards due to the limited availability of ETFs before this year.

We focus exclusively on equity ETFs and apply a strict classification filter to ensure a clean sample. Specifically, only ETFs classified under “Equity Exchange-Traded Fund” (Datastream code: TRAD) and “Exchange Traded” (Datastream code: ET) are included. We exclude all other categories, such as bond ETFs, commodity ETFs, real estate ETFs, mixed asset ETFs, and money market ETFs. Moreover, we do not consider additional classification dimensions such as investment strategy, geographic focus, sector and industry, structure, distribution method, tax and regulatory status, or fund size. ETFs with unclear or missing classifications are also excluded from the final dataset. We address missing values, which are often seen in daily variables like open, high, low, and volume, using a forward-fill method to ensure data completeness and avoid look-ahead bias. Further, we address outliers through winsorization at the two-sided 0.1% level to reduce the influence of extreme values. To avoid survivorship bias, we include both active and inactive stocks and ETFs.

To address potential stale prices in the Datastream daily return index, we identify and remove consecutive observations with unchanged values (forward-filled data). Monthly returns are then computed using the last available non-stale return index within each calendar month. This approach ensures that month-end returns reflect the last valid trading day rather than forward-filled non-trading days.

Table 1 presents the summary statistics of our dataset and provides an overview of the number of observations, number of securities, number of countries, mean, and standard deviation. In panel A, the overall stock return data combines the training,

² The ETF sample was originally constructed through 2022 but extended to 2024 in response to reviewer suggestions, allowing for a longer out-of-sample testing period. It also addresses to test 24 months separately.

Table 1
Summary statistics.

	Periods	# of Obs.	# of securities	# of countries	Mean	Std.	Ave. age	Ave. fee
Panel A: All data								
Stock return	1981.02–2021.05	9,181,979	66,905	67	1.36%	0.21	–	–
ETF return	2005.01–2022.12	573,353	9255	52	0.53%	0.07	79	0.43
Panel B: Train-validate-test split data								
Stock return train	1981.02–2004.12	2,474,549	34,527	52	1.33%	0.21	–	–
ETF return validate	2005.01–2010.12	46,110	1300	29	0.74%	0.10	186	0.47
ETF return test	2011.01–2022.12	527,243	9234	52	0.51%	0.06	78	0.43
ETF return train	2005.01–2010.12	46,110	1300	29	0.74%	0.10	186	0.47
ETF return validate	2011.01–2015.12	127,825	3279	42	0.49%	0.06	144	0.45
ETF return test	2016.01–2022.12	399,418	9234	51	0.52%	0.06	77	0.43

This table summarizes our dataset of both stocks and ETFs. We count the number of observations (# of Obs.), the number of stocks/ETFs (# of Securities), the number of countries that are covered in our dataset (# of Countries), the mean, and standard deviation of the monthly returns of stocks and ETFs. We include the average age in month and fee of ETFs in %.

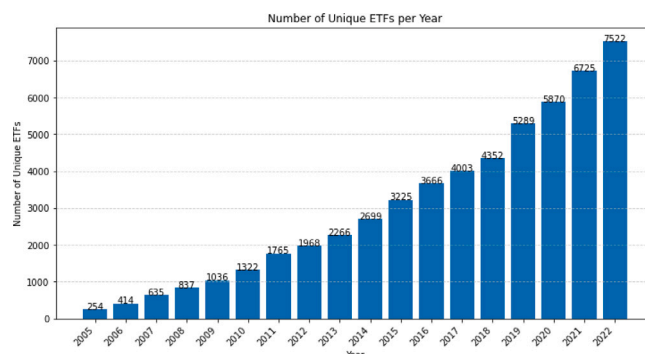


Fig. 1. Number of ETFs by year.

validation, and testing datasets, resulting in a total of 9,181,979 observations. It includes 66,905 securities from 67 countries, with a monthly mean stock return of 1.36% and a standard deviation of 21% from 1981 to 2021. The total equity ETF data consists of 573,353 observations with 9255 securities from 52 countries. The monthly mean of ETF return is 0.53% with a standard deviation of 7% from January 2005 to December 2022. Panel B is structured as a train-validate-test split, dividing the data into distinct periods. These splits are visualized in Fig. 2, which provides a clear timeline of how the stock and ETF data are separated between the training, validation, and test periods. A detailed explanation of the modeling framework is given in Section 2.3. The mean return for the stock training data is 1.33%, with a standard deviation of 21%, while the mean return for ETF training data is 0.74%, with a standard deviation of 10%. The mean returns of ETF validation (test) data are 0.74% (0.51%) and 0.49% (0.52%) for the two training models. Additionally, the average age of ETFs varies between 77 to 186 months, calculated as the time between each ETF's inception date and December 2022. The average net expense ratios range between 0.43% and 0.47% across years.

One important consideration is the distributional difference between stock and ETF returns. While stocks exhibit higher volatility than ETFs, our modeling framework focuses on relative cross-sectional ordering rather than return levels. Moreover, most technical indicators are constructed as normalized transformations of past prices and volumes, making them less sensitive to scale differences in return variance. Consequently, the transfer exercise tests whether structural cross-sectional patterns, rather than return magnitudes, persist from individual securities to diversified baskets.

Additionally, we present figures that illustrate the evolution and composition of the ETF universe in our dataset. Fig. 1 shows a steady increase in the number of unique ETFs from 254 in 2005 to 7522 in 2022, with a substantial growth after 2016. This growth likely influences both the statistical properties of ETF returns and the practical feasibility of trading strategies. Early in the sample period, ETFs are more concentrated in broad market exposures (e.g., SPDR S&P 500 ETF), whereas later years introduce broader geographic coverage and more niche products (e.g., Financial Select Sector SPDR Fund, SPDR Portfolio Developed World ex-US ETF). The relatively small ETF universe in the early years may limit portfolio diversification, particularly in decile formation, which may increase the influence of individual ETFs on results.

Fig. B.1 categorizes ETFs by their geographical investment focus (Datastream code: GFOCUS), which reflects the primary region where at least 50% of assets are allocated or invested, rather than their countries of origin or domicile. The United States dominates (2199 ETFs), followed by Global (1238 ETFs) and China (913 ETFs). Fig. B.2 presents ETF counts by domicile (Datastream code: GEOGN), or legal registration country. The United States dominates (2485 ETFs), followed by Ireland (1428 ETFs), Canada (902 ETFs), Luxembourg, China, and South Korea. Many ETFs are domiciled outside of their investment region.

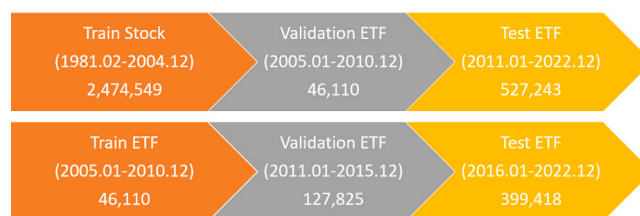


Fig. 2. Train-Test-Split.

2.2. Feature construction: Technical indicators

To initiate our analysis, we employ the Technical Analysis Library (TA) in Python to work with the historical daily price and volume data of the stocks and ETFs.³ TA is a popular open-source library that provides a wide range of technical analysis functions and indicators using time series data. With our comprehensive dataset, we calculate various technical analysis indicators using TA, for example, the average directional movement index (ADX), the relative strength index (RSI), the awesome oscillator (AO), the accumulation/distribution index (ADI), and so on. The indicators are categorized into five groups: momentum indicators, trend indicators, volatility indicators, volume indicators, and other indicators. A comprehensive overview of all 172 technical indicators, categorized into five categories, can be found in Table A.1.

Since many technical indicators are constructed using multi-day price and volume data, we first compute daily technical indicators based on historical daily trading data. To align with our monthly prediction horizon and avoid look-ahead bias, we retain only the values of these indicators on the last trading day of each month, discarding all other daily values. These month-end technical indicators are then matched with the ETF return for the following month, enabling us to examine monthly return predictability.

2.3. Training-validation-test framework

To choose the parameters for our model and mitigate the risk of overfitting, we adopt a training-validation-test framework. In this framework, the training subsample is used for model estimation, the validation subsample is used for hyperparameter tuning, and the testing subsample evaluates the final predictive performance on unseen data, i.e., out-of-sample testing. We use two distinct modeling approaches. In the first case, we train the models on stock data from 1981 to 2004, validate on ETF data from 2005 to 2010, and test on ETF data from 2011 to 2022. This case emphasizes the transferability of predictive signals from stocks to ETFs. In the second case, we train the models on ETF data from 2005 to 2010, validate on ETF data from 2011 to 2015, and test on ETF data from 2016 to 2022. We use this case as a comparison. Fig. 2 summarizes the training-validation-test periods for each case.

The training dataset for stocks (ETF) has a total of 2,474,549 (46,110) observations. The stock training dataset has a rich sample for model estimation compared to the ETF training sample, which is more limited, reflecting the shorter history of the ETF market. When the model is trained on the stock market, the ETF validation and ETF test datasets consist of 46,110 and 527,243 observations, enabling a comprehensive assessment of the model's performance. When the model is trained on the ETF market, the ETF validation and ETF test datasets comprise 127,825 and 399,418 observations (see Fig. 2).

2.4. Predictive modeling and evaluation

First, we train a random forest classification model using the training dataset. The model estimates the monthly probability that an ETF will outperform its peers. For each month, we label ETFs as '1' if their return exceeds the monthly cross-sectional median return and '0' otherwise. Our features are 172 technical indicators, and the targets are 1 or 0.

Then, we conduct hyperparameter tuning using the validation dataset. Specifically, we evaluate different combinations of hyperparameters and select the configuration that achieves the highest validation accuracy. Our candidate hyperparameter grid is based on Table A.5 of Gu et al. (2020), and the selected configurations for each model variant are reported in Table A.2. Fig. 3 illustrates the results of our hyperparameter sensitivity analysis for the two specified model settings. On the x-axis, we show the maximum features used in each model, and on the y-axis, we show the accuracy of the validation data. Each line in the graph corresponds to a different maximum depth level, and the red circles highlight the hyperparameters that perform best for each specific model setting.⁴ We find that simply increasing the model's complexity by increasing the maximum number of features and the model's depth does not necessarily yield better performance, suggesting the importance of hyperparameter sensitivity analysis in avoiding overfitting and guiding model selection. We include 300 trees for both the random forest models.⁵ For the model trained

³ <https://technical-analysis-library-in-python.readthedocs.io/en/latest/>.

⁴ In the random forest model, the "maximum features" parameter limits how many input variables each split considers, introducing randomness that enhances the model's generalization to unseen data, and the "maximum depth" parameter limits how many decision steps each tree can take, helping to balance model complexity and overfitting.

⁵ In the random forest model, the "number of estimators" parameter determines how many individual decision trees the forest will contain. It is also called "number of trees".

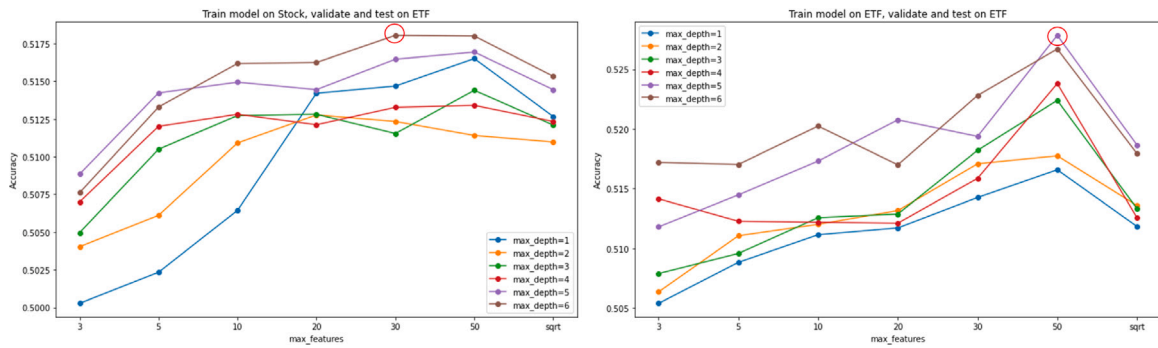


Fig. 3. Hyperparameter sensitivity analysis.

on stocks and validated on ETFs, the optimal settings are a maximum depth of 6 and 30 maximum features. Conversely, for the model trained and validated on ETFs, the ideal parameters are a maximum depth of 5 and 50 maximum features.

Finally, during the prediction phase on the testing dataset, each decision tree in the random forest outputs a class probability, and the final probability of outperformance is obtained by averaging these probabilities across all trees. This probabilistic output forms the basis for ranking ETFs and constructing portfolios. Based on the predicted outperformance probabilities from the random forest model, we construct long–short portfolios that are rebalanced monthly. For each month, we sort ETFs by their predicted probabilities and take equal-weighted long positions in the top decile and short positions in the bottom decile; we apply the same approach to stock portfolios. To assess the robustness of our findings, we also implement alternative machine learning models – logistic regression, decision tree classification (DT), gradient-boosted decision trees (GBDT), and XGBoost classification – as additional robustness checks to ensure model stability. Table A.3 consolidates the full modeling and evaluation process, from data preparation through hyperparameter tuning to portfolio construction.

3. Empirical results

3.1. Return predictability

We next present the results based on models trained with technical indicators from the global stock and ETF markets, using the selected hyperparameters. Table 2 provides empirical insights into the predictive potential of technical signals sourced from both the global stock market and the global ETF market with monthly rebalancing. The left part of the table displays results obtained from models trained on stock data spanning the years 1981 to 2004. The right part of the table shows results from models trained on ETF data spanning the years 2005 to 2010.

When the model is trained on stock indicators, we observe a substantial monthly average long–short return of 0.76% in local currency, with high statistical significance (t -statistic = 2.76), and 0.73% in USD, with similar statistical significance (t -statistic = 2.78). Then, we adjust the returns using three different models: the CAPM model, which includes the market risk factor; the Fama–French 3-factor model, which includes the market risk factor, the size risk factor, and the value risk factor; and the Carhart 4-factor model, which extends the 3-factor model by adding the momentum factor. For simplicity, we do not consider country-specific risk factors and instead use value-weighted world factors. The world factors are constructed by aggregating country-level Fama–French–style factors using market capitalizations (in USD) from Datastream as weights. After adjusting the market risks, the model yields an average long–short return of 0.62% in local currency with a t -statistic of 2.27. As throughout the paper, these risk-adjusted returns are reported to assess the strength and robustness of cross-sectional return rankings rather than to claim economically scalable abnormal returns. The risk-adjusted long–short returns after the Fama–French three-factor model and the Carhart four-factor model are 0.60% (t -statistic = 2.19) and 0.59% (t -statistic = 2.00) in local currency, respectively. Adjusting the returns in USD using the corresponding world risk factors yields similar alphas, confirming that the results are robust to currency denomination, namely 0.64% (t -statistic = 2.46) after market adjustment, 0.63% (t -statistic = 2.40) after Fama–French, and 0.62% (t -statistic = 2.20) after Carhart. For the model trained on ETF technical indicators, we find a somewhat lower raw return of 0.58% in local currency with less statistical significance (t -statistic = 1.84). To provide further insight, Fig. B.3 presents average returns and alphas across all ten decile portfolios when the model is trained on stock indicators, which allows us to assess the monotonicity of the return pattern.

While the ETF-trained model does not outperform the stock-trained model in general, it is important to note that the evaluation periods for the two models are different. The evaluation period for the ETF-trained model is shorter than for the stock-trained model due to the shorter history of the ETF market. To enhance comparability, we further evaluate the stock-trained model over the same period as the ETF-trained model (2016–2022). During this period, the stock-trained model achieves a long–short return of 0.64% with a t -statistic of 1.96. Compared to the ETF-trained model, the stock-trained model continues to demonstrate stronger performance. Even though the performance difference is modest in this overlapping period, it is important to recognize that the stock-trained model allows prediction across a much longer horizon, beginning in 2005. These findings highlight the importance and

Table 2
One-month predictability.

Trained on stocks						Trained on ETFs		
Local currency			USD			Local currency		
Long	Short	Long-short	Long	Short	Long-short	Long	Short	Long-short
Panel A: ETF raw return								
1.02%	0.26%	0.76%*** (2.76)	0.93%	0.20%	0.73%*** (2.78)	0.98%	0.40%	0.58%* (1.84)
Panel B: ETF CAPM Alpha								
0.29%	−0.33%	0.62%** (2.27)	0.25%	−0.40%	0.64%** (2.46)	0.27%	−0.14%	0.41% (1.34)
Panel C: ETF 3 Factor Alpha								
0.31%	−0.29%	0.60%** (2.19)	0.25%	−0.37%	0.63%** (2.40)	0.31%	−0.10%	0.41% (1.34)
Panel D: ETF 4 Factor Alpha								
0.43%	−0.17%	0.59%** (2.00)	0.34%	−0.28%	0.62%** (2.20)	0.41%	−0.19%	0.60%* (1.89)
Panel E: Size Neutral ETF raw return								
0.37%	−0.12%	0.49%* (1.74)	0.35%	−0.10%	0.45% (1.60)	0.77%	0.54%	0.23% (0.70)
Panel F: Style Neutral ETF raw return								
0.41%	−0.16%	0.57%** (2.14)	0.38%	−0.18%	0.56%** (2.14)	0.77%	0.54%	0.23% (1.02)

This table reports ETF predictability results for two different training scenarios: one with models trained on stock data (1981–2004) and the other with models trained on ETF data (2005–2010). The evaluation period spans 2011–2022 for the stock-trained model and 2016–2022 for the ETF-trained model. The table is divided into six panels. Panel A shows the evaluation of these models when applied to ETF raw return predictions, displaying metrics such as long return, short return, long-short mean return, and mean return. Panels B, C, and D present the evaluation results for risk-adjusted ETF raw returns. Panel E shows the evaluation of size-adjusted long return, short return, long-short mean return, and mean return. Panel F shows the evaluation of style-adjusted long return, short return, long-short mean return, and mean return. For results reported in local currency, the corresponding risk factors are also measured in local currency, whereas for results reported in USD, risk factors are measured in USD. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

robustness of the stock-trained model across different evaluation periods and highlight its effectiveness in predicting cross-sectional ETF performance as a proxy for basket-level returns.

Besides adjusting for raw returns in Table 2 Panels B to D, we consider the size and style effects in our analysis of long and short portfolios. The adjustment is crucial because the long or short portfolios may predominantly comprise ETFs that invest in large-cap stocks or value stocks, potentially biasing the results. To neutralize the size and style effect, we collect the Morningstar equity style box, which categorizes funds into nine distinct categories based on size (small, mid, and large) and style (growth, blend, and value). We adjust the returns by subtracting the mean returns of ETFs within each size and style category before calculating the overall portfolio return. The adjusted size-neutral and style-neutral returns are presented in Table 2 Panels E and F. We observe that the style-neutral return 0.57% with a *t*-statistic of 2.14 is higher than the size-neutral return 0.49% with a *t*-statistic of 1.74, indicating that the cross-sectional return rankings are more strongly associated with style-related exposures than with firm size; nevertheless, they are still statistically significant. Table A.4 reports the results when we restrict our training sample to larger stocks, defined as the market value larger than the median, given the hypothesis that these stocks are more likely to be held by ETFs.

Fig. 4 shows the cumulative raw return of the long-short portfolio in the ETF market from the year 2011 to 2022, with an extension through 2024 (shaded in gray). We use the model trained on the global stock dataset. The cumulative return curve starts close to zero and steadily increases, reaching roughly 1.5 (+150%) at the end of 2022; during the extended period (2023–2024), it continues upward, approaching 3 (+300%). This cumulative pattern illustrates the persistence of the cross-sectional return spreads generated by the stock-trained model. Consequently, in the following analysis, we focus on the global stock-trained model.

We next examine the long-term horizon by increasing the rebalance periods. In Fig. 5, we illustrate the outcomes of portfolio rebalancing at different time intervals. From left to right, the graphs represent portfolio rebalancing frequencies of 6, 12, 24, and 36 months, respectively. We show that robust predictability is primarily confined to the first month and decreases in the subsequent months in the ETF markets. This phenomenon indicates the presence of predictability in the short term while indicating the absence of predictability in the long term for ETF performance.

3.2. Indicator importance

To better understand the contribution of different types of predictors to ETF ranking performance, we conduct a “feature importance” analysis by grouping the features into five categories as described in Section 2.2. The grouped feature importance from the random forest classification model is as follows: With a relative importance of 0.36, the volatility group emphasizes the greatest significance. Volatility measures, such as average true range or bollinger bands, contribute significantly to the model’s



Fig. 4. Cumulative ETF returns over time.

Notes: The shaded gray area indicates the most recent period used for out-of-sample evaluation. Cumulative returns are computed from equal-weighted ETF portfolios between 2011 and 2024.

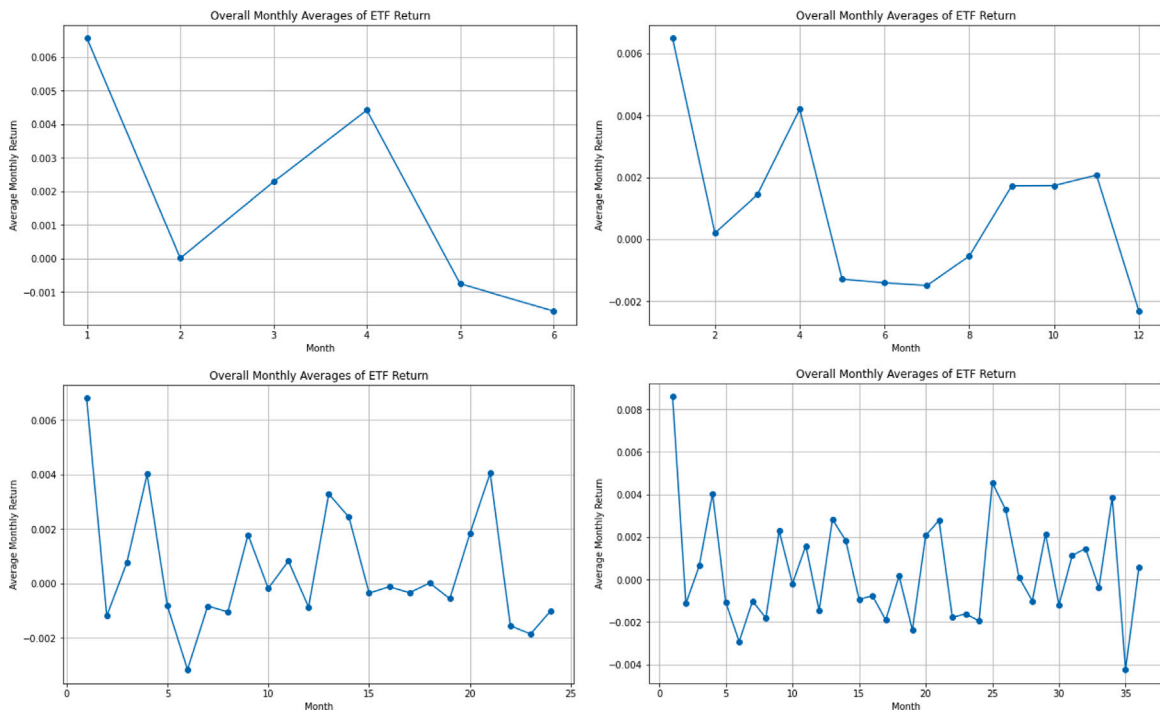


Fig. 5. Different rebalancing periods.

ability to classify and predict outcomes. The trend group, with a relative importance of 0.25, ranks as the second most significant, following the volatility group. Trend indicators, such as exponential moving average or moving average convergence divergence, provide insights into the direction and strength of price movements. The momentum group of features has a relative importance of approximately 0.25 and exhibits a comparable influence on the classification model to the trend group. It suggests that the recent price movements and trends of the securities are influential factors. The volume group has the lowest relative importance of around 0.02, indicating that volume-related indicators have a relatively minor influence on the classification model. However, they still contribute to some extent in assessing predictability. The other group has a relative importance of 0.11, indicating that other miscellaneous factors contribute to the classification model (see Fig. 6).

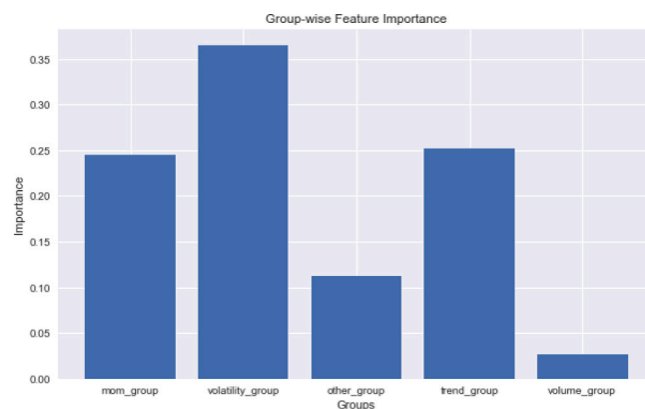


Fig. 6. Grouped feature importance.

Table 3

Predictive performance by indicator category.

Indicators	n_estimators	max_depth	max_features	Long	Short	Long-short
Momentum	300	6	20	0.91%	0.26%	0.66%*** (2.57)
Trend	300	115	“sqrt”	0.72%	0.43%	0.29% (1.55)
Volatility	300	3	“sqrt”	0.90%	0.26%	0.64%** (2.23)
Volume	300	225	“sqrt”	0.64%	0.60%	0.04% (0.25)
Others	300	4	3	0.80%	0.19%	0.61%** (2.15)

This table reports the classification results when we restrict our training data to different categories of stock technical indicators. The models are trained from 1981 until 2004 and evaluated from 2011 until 2022. The evaluation period spans 2011–2022 for the stock-trained model. The hyperparameter selection follows Gu et al. (2020) and is tuned accordingly. The models are evaluated on the ETF market. “sqrt” indicates that the number of features considered at each split is the square root of the total number of available features. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

The default feature importance in random forest models, measured by the mean decrease in impurity (MDI) during training, captures the extent to which the model relies on each feature to reduce classification error (Breiman, 2001; Pedregosa et al., 2011). However, it does not necessarily reflect the feature’s ability to generalize to unseen data. This limitation becomes especially relevant when there is a distribution shift between training and testing datasets.⁶ Thus, the results in Fig. 6 should be interpreted with caution, as they do not guarantee predictive relevance on the test set.

To address this issue, we further assess feature contributions in the test data by training separate models using each group of stock technical indicators and subsequently validating and testing them in the ETF market. The empirical findings, presented in Table 3, examine the predictive power of each indicator category. We find that momentum, volatility, and other indicators yield significant long–short returns (0.66%, 0.64%, and 0.61%, respectively), while trend and volume indicators show weaker performance. All single-group models underperform the full model, which achieves a 0.76% long–short return (see Table 2). To determine whether the weaker performance of trend and volume indicators comes from poor feature quality or over-simplistic models, we revisit the hyperparameter tuning process. The default grid from Gu et al. (2020) selects shallow trees (max depth = 1), leading to low predictive confidence and ineffective portfolio rankings. We increase the model complexity (e.g., max depth, maximum features) to mitigate underfitting. Despite this, trend and volume indicators remain weak, reinforcing that the limitation lies in the features themselves rather than the model setup.

We confirm these results using logistic regression, a linear model. The results, detailed in Table A.5, indicate that models trained on momentum indicators show a 0.58% long–short return with high significance (t -statistic = 2.98). Conversely, models trained on volume indicators even exhibit a negative average long–short return at –0.28%. Although the model trained on trend indicators shows a positive long–short return, this is not statistically significant.

Our results that momentum and volatility indicators have the highest predictive importance are consistent with the literature. For example, Hull and Qiao (2017) document that technical momentum signals and volatility-related measures are strong predictors of SPDR S&P 500 ETF Trust (SPY). More specifically, Li et al. (2019) document robust momentum profits in ETFs, showing that

⁶ In our study, this distribution shift arises because the model is trained on stock data but evaluated on ETF data, which follow different distributions.

Table 4
Model evaluation across geographical regions.

Geographical focus	Long	Short	Long–short	CAPM Alpha	3 Factor Alpha	4 Factor Alpha
Global	1.07%	0.41%	0.66%** (2.35)	0.53%* (1.87)	0.51%* (1.80)	0.46% (1.50)
Europe	1.08%	0.35%	0.73%*** (2.60)	0.62%** (2.20)	0.60%** (2.12)	0.57%* (1.87)
US	0.82%	0.34%	0.48% (1.60)	0.38% (1.27)	0.37% (1.23)	0.52% (1.59)
China	1.07%	0.20%	0.87%*** (3.92)	0.86%*** (3.80)	0.85%*** (3.73)	0.80%*** (3.25)
Korea	1.05%	0.36%	0.69%** (2.44)	0.57%** (2.00)	0.55%* (1.92)	0.52%* (1.67)
Global Ex US	1.08%	0.36%	0.72%** (2.57)	0.60%** (2.15)	0.59%** (2.08)	0.56%* (1.82)

This table reports the results from the global stock-trained model testing on the different geographical focuses from 2011 until 2022. We show, for different investment areas which have at least 300 ETFs, the long return, short return, and long–short return with statistical significance. For all regions, risk factors are measured in local currency. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

past return patterns strongly predict future ETF performance. This suggests that the predictive signals identified in our model are economically meaningful in the sense of improving cross-sectional return rankings.

3.3. Different geographic investment focus

Using the previous global stock-trained model, we conduct an analysis of predictability across various markets. We categorize the ETFs based on their geographic investment focus. Our dataset comprises information on the geographic investment focus of 7639 ETFs, sourced from Datastream. In Fig. B.1 we present the number of ETFs in each geographic investment focus. For our analysis, we focus on regions with more than 300 ETFs to ensure a substantial sample size. The majority of ETFs in our dataset invest in the U.S. market. As an exception, we also include the Global ex-U.S. category, even though it contains fewer than 300 ETFs, because it provides a useful complementary benchmark to the U.S. segment.

Table 4 presents the evaluation results of ETF markets based on different geographic investment focuses. The evaluation includes ETFs that target global stocks, European stocks, US stocks, Chinese stocks, Korean stocks, and global stocks excluding US stocks. Using the global stock-trained model to generate predictions across geographic segments, we observe that long–short return spreads are positive across all regions, although their magnitudes and statistical significance differ. Notably, ETFs focusing on the Chinese market exhibit the strongest predictive performance, with a long–short return of 0.87% (t -statistic = 3.92). In contrast, ETFs concentrating on the U.S. market exhibit weaker predictive performance, with a long–short return of 0.48% (t -statistic = 1.60). The table also reports the risk-adjusted alphas in each region. These findings suggest that the geographic focus of ETFs plays an important role in the strength of cross-sectional return predictability, with Chinese-focused ETFs exhibiting stronger cross-sectional return rankings than U.S.-focused ETFs. This observation aligns with the notion that the strength of basket-level return predictability is influenced by the efficiency of the underlying stock market. In more efficient markets (e.g., the U.S.), where information is incorporated into prices more rapidly, the persistence of cross-sectional return rankings tends to be lower, whereas in less efficient markets (e.g., China), slower information incorporation is associated with stronger short-horizon predictability.

In addition to evaluating monthly predictability, we also examine the cumulative returns for each investment area over time. As depicted in Fig. 7, initially, the cumulative returns for each investment area show little disparity. However, as time passes, ETFs investing in the Chinese market exhibit more persistent cumulative long–short return spreads compared to ETFs focused on the U.S. market. The growing gap in cumulative return spreads over time reflects differences in the persistence of predictive signals across markets. These findings underscore the importance of accounting for market efficiency when evaluating the strength and persistence of predictive signals in ETF rankings.

3.4. Implementation considerations

3.4.1. Portfolio turnover analysis

To complement the return-based evaluation of our predictive strategies, we analyze the turnover rates related to each portfolio. Turnover provides insights into the degree of trading activity and implementability of each strategy. While most of the academic papers use value-weighted or dollar-based turnover formulas for stock portfolios (e.g., Brandt et al. 2009, Gu et al. 2020), we calculate a count-based turnover measure, which is the average number of ETFs entering or exiting a portfolio between consecutive months, normalized by the portfolio size. This approach is consistent with earlier work in momentum strategies (e.g., Jegadeesh and Titman 1993), and this measure is appropriate for our setting, because our portfolios are equally weighted. Especially, for each side (long or short), turnover is defined as the number of added and removed ETFs divided by twice the number of ETFs in the portfolio. The total turnover is calculated as the sum of the long and short turnover rates:

$$\text{Turnover}_t^{(long)} = \frac{|\text{Added}_t^{(long)}| + |\text{Removed}_t^{(long)}|}{2N}$$

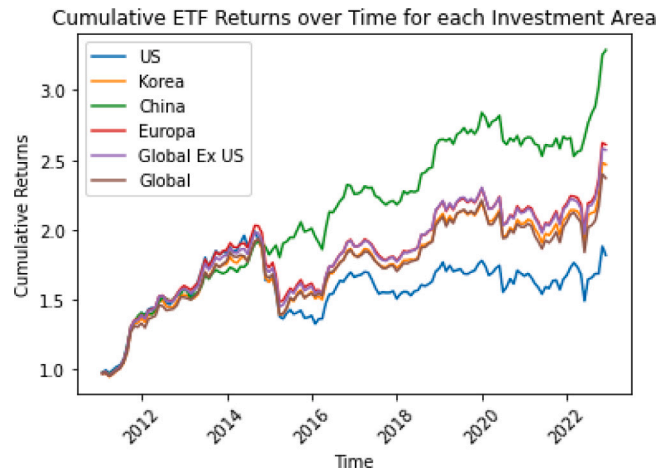


Fig. 7. Cumulative ETF returns over time in different areas.

$$\text{Turnover}_t^{(short)} = \frac{|\text{Added}_t^{(short)}| + |\text{Removed}_t^{(short)}|}{2N}$$

$$\text{Total Turnover}_t = \text{Turnover}_t^{(long)} + \text{Turnover}_t^{(short)}$$

where $|\text{Added}_t^{(long,short)}|$ is the number of new ETFs added to the portfolio at time t . $|\text{Removed}_t^{(long,short)}|$ is the number of ETFs removed from the portfolio at time t . N is the number of ETFs in the portfolio. For example, if the strategy selects the top 10% of all ETFs each month and there are 120 ETFs in total, then $N = 12$, which means the portfolio will include the 12 highest-ranked ETFs based on the model signal.

Table 5 reports the average monthly turnover rates for both the long and short portfolios in different strategy settings: the baseline (full replacement) and three sticky strategies using thresholds of 40%, 50%, and 60%. The sticky threshold determines how much of the current portfolio is allowed to persist into the next period. For example, a 40% threshold means that ETFs from the portfolio of the previous period are retained in the current portfolio if they remain within the top (or bottom) 40% of the universe, reducing unnecessary trading. The remaining positions are filled by new ETFs ranked at the top or bottom based on the model signals.

The baseline strategy, which fully replaces portfolios each month, yields the highest turnover, with an average of 87.52% on the long side and 80.05% on the short side, resulting in a total turnover of 167.57%. As we increase the sticky threshold, turnover decreases substantially. For example, under the sticky strategy with a 50% threshold, turnover falls to 49.32% for both long and short portfolios, with a total turnover of 98.64%. When using a 60% threshold, the total turnover drops even further to 78.67%.

Reducing turnover does not materially weaken the model's cross-sectional ranking performance. Despite relaxing the need for full portfolio rebalancing, the long–short return spreads remain statistically significant. For example, the specification based on a 50% threshold yields an average monthly long–short return spread of 0.51% with a t -statistic of 2.08. The corresponding CAPM-adjusted return spread is 0.44% with a t -statistic of 1.76. Even with a more stringent 60% threshold, the predictive signal remains economically meaningful, with an average long–short return spread of 0.46% (t -statistic = 1.90). These results show that substantial reductions in trading activity, dropping from 167.57% under full replacement to 78.67%, can be achieved without materially weakening the strength of the cross-sectional return rankings. In general, this is the trade-off between portfolio stability and responsiveness. Although sticky portfolios help reduce turnover and may lower transaction costs, the associated return spreads remain statistically distinguishable from zero.

To provide a simple and transparent assessment of how transaction costs would affect the reported return spreads, we conduct a back-of-the-envelope break-even transaction cost analysis following the approach used in related studies (e.g., Azevedo and Hoegner, 2023). The maximum round-trip transaction cost that would still leave the return spread statistically significant at the 5% level can then be approximated as:

$$\text{RTC}_{5\%} = 2 \left(1 - \frac{1.96}{T} \right) \frac{\bar{\mu}}{\overline{TO}},$$

where $\bar{\mu}$ denotes the average monthly long–short return spread and T its corresponding t -statistic. $\overline{TO}$ is the average one-sided monthly portfolio turnover, computed as half of the total turnover reported in Table 5. The leading factor of two converts one-sided trading costs into round-trip costs. This measure should be interpreted strictly as a break-even benchmark and not as an estimate of achievable transaction costs or net trading performance.

Applying this calculation to the full-replacement strategy, where a total monthly turnover is 167.57% and a t -statistic is 2.76, yields an implied break-even round-trip cost of approximately 53 basis points. Economically, this means that if the effective cost of entering and exiting a position exceeds 0.53% per round trip, the corresponding return spread would no longer be statistically significant at the 5% level. In contrast, for the sticky rebalancing strategy with a 50% retention threshold, where total turnover

Table 5

Portfolio returns and turnover: Baseline vs. sticky strategy.

Strategy	Long	Short	Long–short	CAPM Alpha	3 Factor Alpha	4 Factor Alpha	Long TO	Short TO	Total TO
Baseline (Full replacement)	1.02%	0.26%	0.76%*** (2.76)	0.62%** (2.27)	0.60%** (2.19)	0.59%** (2.00)	87.52%	80.05%	167.57%
Sticky (40% threshold)	0.86%	0.28%	0.58%** (2.34)	0.50%** (1.97)	0.48%* (1.89)	0.40%** (1.47)	59.41%	59.11%	118.52%
Sticky (50% threshold)	0.80%	0.29%	0.51%** (2.08)	0.44%* (1.76)	0.42%* (1.69)	0.34% (1.24)	49.32%	49.32%	98.64%
Sticky (60% threshold)	0.76%	0.30%	0.46%* (1.90)	0.40% (1.62)	0.38% (1.55)	0.34% (1.26)	39.35%	39.33%	78.67%

This table reports the long and short portfolio returns, long–short (l–s) mean returns with corresponding *t*-statistics (in parentheses), and turnover (TO) figures for both long and short positions, as well as total turnover, which is the sum of long turnover and short turnover. Results are shown for a baseline strategy with full replacement and for sticky strategies with varying retention thresholds (40%, 50%, 60%). ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The evaluation period spans from 2011 to 2022.

declines to 98.64%, and the return spread remains statistically significant (*t*-statistic = 2.08), the implied break-even round-trip cost decreases to approximately 12 basis points. This means transaction costs exceeding 0.12% per round trip would be sufficient to eliminate statistical significance. For the most conservative sticky strategy with a 60% retention threshold, the return spread is no longer statistically significant at the 5% level, and the corresponding break-even bound calculation is not possible. These calculations are intended as a back-of-the-envelope break-even benchmark based on turnover and are not meant to represent realistic net performance, because we still ignore implementability, capacity constraints, and market impact. Instead, they provide a transparent scaling reference, reinforcing that the reported return spreads should be interpreted as gross-of-cost evidence of statistical predictability and cross-asset signal transfer rather than economically attainable trading profits.

3.4.2. Liquidity effect

The liquidity of exchange-traded funds (ETFs) may play an important role in return predictability and portfolio performance. It may be anticipated that ETFs generally offer superior liquidity compared to their underlying securities, largely due to their intraday tradability and efficient arbitrage mechanisms. However, as highlighted by Box et al. (2021), this liquidity advantage is not universal. Their findings indicate that the bid–ask spreads of the ETF are narrower than those of the underlying securities in only 45% of the cases, suggesting substantial variability in the liquidity of the ETF. Importantly, their analysis shows that ETF trading predominantly reflects fundamental movements in the underlying portfolio rather than amplifying nonfundamental shocks. This suggests that stock-based return predictors remain highly relevant, as ETFs closely mirror the price changes of their underlying assets.

To quantify liquidity differences, we compute the relative bid–ask spread as follows:

$$\text{Relative Spread}_{i,t} = \frac{P_{ask,i,t} - P_{bid,i,t}}{\frac{(P_{ask,i,t} + P_{bid,i,t})}{2}}$$

where the Relative Spread for asset *i* on day *t* is calculated as the difference between the local ask price ($P_{ask,i,t}$) and the local bid price ($P_{bid,i,t}$), divided by their midpoint price. To assess liquidity differences over time, we first compute daily relative spreads, then aggregate them monthly by taking the median across all available observations within each calendar month, separately for ETFs and individual stocks.

Figs. B.4–B.6 illustrate the liquidity in different asset classes and frequencies. Fig. B.4 compares the monthly relative bid–ask spread of ETFs and stocks. This reveals that stocks consistently show higher relative spreads (lower liquidity) with notable spikes during the 2008 financial crisis. Fig. B.5 highlights the differences between ETFs and highly liquid stocks, which are defined as those with market capitalizations above the cross-sectional median. This reveals that ETFs tend to have higher spreads during market distress periods, reflecting temporary liquidity frictions. Fig. B.6 compares daily and monthly ETF spreads.

We examine how ETF market liquidity affects the strength of the cross-sectional return rankings generated by our global stock-based model from Section 3.1 without retraining the model, we apply it to ETFs grouped by high and low liquidity and evaluate the resulting long–short return spreads to assess whether liquidity differences affect the strength of stock-based predictive signals. The results in Table 6 show that predictive performance is stronger among ETFs with lower liquidity. The monthly long–short return spread is 0.91% (*t*-statistic = 2.31) for low-liquidity ETFs, compared with 0.52% (*t*-statistic = 1.29) for high-liquidity ETFs. Consistent with market efficiency theories, higher liquidity appears to accelerate price adjustment, reducing the persistence of return spreads. Conversely, lower liquidity is associated with slower information incorporation, which is reflected in stronger short-horizon cross-sectional return rankings. Thus, liquidity frictions appear to be an important determinant of the strength and persistence of return predictability in ETF rankings.

4. Further robustness tests

4.1. Various models

In the previous section, our analysis primarily focuses on the implementation of the random forest classification model. Now we aim to broaden our analytical framework by incorporating other alternative models. This expansion includes not only different

Table 6
High liquid ETF vs. low liquid ETF.

Classification	High liquid ETF			Low liquid ETF		
	Long	Short	Long–short	Long	Short	Long–short
Relative spread	0.68%	0.16%	0.52% (1.29)	0.41%	−0.50%	0.91%** (2.31)

This table reports the performance of long–short strategies for ETF returns, evaluated using the global stock-based predictive model. The model is applied separately to high-liquid and low-liquid ETFs, where liquidity is classified based on relative spread. “Long ret” and “short ret” represent the average monthly returns of the long and short portfolios, respectively. “l–s mean” denotes the mean monthly return of the long–short strategy, with associated *t*-statistics in parentheses. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The evaluation period spans from 2011 to 2022.

Table 7
Other model settings with validation.

Model	Classification based			Regression based		
	Long	Short	Long–short	Long	Short	Long–short
Logistic regression	0.86%	0.19%	0.67%*** (3.16)	0.53%	0.40%	0.13% (0.53)
Decision tree	0.90%	0.26%	0.64%** (2.48)	0.76%	0.50%	0.26% (1.37)
GBDT/GBRT	0.74%	0.37%	0.37%** (2.28)	–	–	–
XGBoost	0.96	0.13	0.83%*** (3.36)	0.62%	0.30%	0.32% (1.59)

This table reports a robustness check for our global stock-trained model. We include classification-based and regression-based models. The classification-based models are linear model (logit model), decision tree, gradient boosting decision tree, and XGBoost, and the regression-based models are linear model (ordinary least square), decision tree, gradient boosting regression tree, and XGBoost. We evaluate the model on ETF markets and show the long return, short return, and long–short return with statistical significance. The ‘–’ symbol is used to indicate the absence of results for the Gradient Boosting Regression Tree (GBRT) model due to computational time requirements that made it infeasible to obtain. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively. The evaluation period spans from 2011 to 2022.

classification-based models, such as Linear Models, Decision Trees (DT), Gradient Boosting Decision Trees (GBDT), and XGBoost, but also regression-based models. By applying different models, we seek to improve the comprehensiveness and resilience of our analysis. The results obtained from this extended analysis, using the same training and testing data as presented in Fig. 2 (training data in the stock market, and testing data in the ETF market), are outlined in Table 7.

Table 7 left part shows the evaluation from the classification-based model. The XGBoost model exhibits the strongest cross-sectional ranking performance among the classification models in the ETF market. This model generates a long–short return spread of 0.83%, reflecting a clear separation between top- and bottom-ranked ETFs. This indicates a strong ability to distinguish relative ETF performance in a cross-sectional setting. On the right side of Table 7, regression models trained on stocks generate relatively lower returns in the ETF market compared with the classification models. This underperformance of regression models finds support in a recent study by Breitung (2023), which underscores the propensity of regression models to place disproportionate emphasis on stocks with the most extreme return behaviors during their training process. This overfitting, combined with regression models’ weaker generalization, could explain this return difference. The presence of “–” symbols in the table indicates the absence of results for the GBRT model. This exclusion is due to the GBRT model’s extensive computational time, making its use impractical and infeasible for our study. It is important to note that regression-based models need significantly longer execution times when compared to their classification-based counterparts. In our analysis, to implement the long–short trading strategy, our target is the ranking of the ETFs based on the outperformance probability, and the exact return prediction is not necessary. In light of this, we prioritize the use of classification-based models for our analysis.

4.2. Test on bond ETF market

For a further robustness test, we collect bond ETF data from Datastream and apply the global stock-trained model to make predictions on the bond ETF market. Applying both regression-based and classification-based models using stock technical indicators, we find that cross-sectional return rankings do not transfer well to the bond ETF market. The inherent distinctions between global stocks and bond ETFs, whose underlying assets consist of corporate and government bonds, likely limit the transferability of stock-based predictive signals.

Table 8 presents the performance evaluation of various models when tested on the bond ETF market. In the classification-based category, the decision tree model exhibits the strongest cross-sectional return ranking performance, with a long–short return spread

Table 8
Bond ETF market.

Model	Classification based			Regression based		
	Long	Short	Long–short	Long	Short	Long–short
Logistic regression	0.08%	0.13%	–0.05% (–0.38)	0.44%	0.06%	0.38% (1.16)
Decision tree	0.37%	–0.03%	0.40%** (2.28)	0.04%	0.02%	0.02% (0.26)
Random forest	0.16%	0.28%	–0.12% (–0.33)	0.24%	0.06%	0.18% (1.19)
GBDT/GBRT	0.08%	0.13%	–0.05% (–0.38)	0.17%	0.07%	0.10% (1.22)
XGBoost	0.06%	0.03%	0.03% (0.39)	0.18%	0.10%	0.08% (0.68)

This table reports a robustness check on the bond ETF market for our global stock-trained model. We include classification-based and regression-based models. The classification-based models are linear model (logit model), decision tree, gradient boosting decision tree, and XGBoost, and the regression-based models are linear model (ordinary least square), decision tree, gradient boosting regression tree, and XGBoost. We evaluate the model on bond ETF markets and show the long return, short return, and long–short return with statistical significance. The evaluation period spans from 2011 to 2022.

of 0.40% (t -statistic = 2.28). Besides, models such as the linear model, random forest, GBDT, and XGBoost display a range of performance levels, yielding average long–short returns that span from –0.12% to 0.03% without statistical significance. These returns are notably lower when compared to the long–short returns previously generated in the equity ETF market (see [Tables 2 and 7](#)). In the “regression-based” category, the decision tree model shows a low average long–short return of 0.02% and no statistical significance. The random forest model demonstrates a somewhat stronger performance with a return of 0.18%, but also not significant. The weak transfer of stock-based signals to bond ETFs indicates that the model captures equity-specific cross-sectional return dynamics rather than universal technical trading patterns. This finding reinforces the interpretation that cross-asset transfer requires structural similarity between the training and testing assets.

Overall, the findings indicate that models trained on stock data do not yield strong cross-sectional return rankings when applied to the bond ETF market, and it requires tailored modeling approaches to effectively capture its specific characteristics. It could be that models trained on corporate bond and government bond data, as opposed to stock data, might be more effective in the bond ETF market.

4.3. A true out of sample test

To further evaluate the robustness of our predictive framework, we extend our analysis into a recent out-of-sample period covering 2023–2024 (24 months).⁷ We apply the previously trained global stock-based model to this shorter evaluation period without any adjustments, and the long–short strategy yields a monthly mean return of approximately 1.31% with a t -statistic of 1.44. This decline in statistical significance comes from the relatively short evaluation horizon. Nonetheless, the persistence of positive return spreads during this recent period suggests that the cross-sectional return rankings remain informative, despite the short evaluation window. When adjusting for market risk using the CAPM model, the risk-adjusted monthly return is 0.90% with a t -statistic of 0.86. Under the Fama-French three-factor model, the adjusted return rises to 1.25% (t -stat = 1.00). Moreover, when accounting for the momentum factor from the Carhart four-factor model, the adjusted return further increases to 1.55% with an improved t -statistic of 1.21, despite only investigating 24 months.

Combining the full out-of-sample period from 2011 to 2024, the long–short return spread averages 0.84% per month with a t -statistic of 3.11. This result highlights the robustness of the cross-sectional return rankings generated by the global stock-based model across a prolonged out-of-sample horizon. Adjusting for market risk using the CAPM model, the strategy continues to generate a statistically significant alpha of 0.69% per month (t -statistic = 2.54). When controlling for size and value factors from the Fama-French three-factor model, the alpha remains stable at 0.61% with a t -statistic of 2.26. In particular, incorporating the momentum factor from the Carhart four-factor model yields an even higher risk-adjusted return of 0.50% with the t -statistic of 1.75. These results affirm the stability of the cross-sectional return rankings and their robustness to standard risk factor adjustments. All results are reported in [Table A.6](#).

4.4. Quintile- and vigintile-based portfolio construction

To further assess the robustness of our main findings, we construct portfolios based on quintile/vigintile rankings of the predicted outperformance probabilities (2011.01–2022.12). [Tables A.7 and A.8](#) extend this analysis to quintile-based and vigintile-based

⁷ We thank an anonymous referee for making this suggestion.

sorting, where ETFs are grouped into five equally sized portfolios and twenty equally sized portfolios instead of ten. The results confirm the robustness of the cross-sectional ranking approach. The long–short raw return, based on Q5–Q1, is 0.66% (t -statistic = 3.07), and based on Q20–Q1, is 0.88% (t -statistic = 2.67). The risk-adjusted performance under CAPM, 3-Factor, and 4-Factor models yields long–short alphas of 0.56% (t -statistic = 2.62), 0.55% (t -statistic = 2.54), and 0.53% (t -statistic = 2.27), respectively for the quintile based ranking, and 0.70% (t -statistic = 2.14), 0.68% (t -statistic = 2.07), and 0.66% (t -statistic = 1.86), respectively for the vigintile based ranking. These findings suggest that alternative portfolio partition schemes yield consistent return spreads and ranking patterns relative to the decile-based approach.

5. Conclusion

This paper demonstrates that technical indicators originally developed for stock selection contain informative signals for ranking future ETF performance, particularly when ETFs have relatively short return histories. First, we document short-horizon cross-sectional return predictability in the international ETF market using the global stock-trained model, indicating that stock-based signals are informative for monthly cross-sectional rankings. The strength of the cross-sectional return rankings is largely preserved even when portfolio turnover is reduced using sticky rebalancing schemes. This suggests that the predictive signals are not mechanically driven by excessive rebalancing and remain stable under alternative portfolio construction constraints. Second, feature importance analysis from the random forest classification model shows that volatility, trend, and momentum indicators contribute most strongly to cross-sectional return rankings when the model is trained on stock data. Our grouped-indicator analysis confirms that volatility and momentum signals account for most of the observed predictability, while trend and volume indicators are less informative when used in isolation. Third, our findings suggest that cross-sectional return rankings are stronger among lower-liquidity ETFs, consistent with slower incorporation of information into prices in the presence of trading frictions. Finally, our findings indicate that ETFs tracking more efficient stock markets exhibit weaker short-horizon predictability. This suggests that in markets where information is rapidly incorporated into prices, historical technical indicators contain less incremental information for cross-sectional return ranking.

Based on our findings, future research could examine cross-asset signal transfer in a broader context, extending it beyond equity ETFs to other asset classes that are structurally linked to the underlying securities. For example, the technical indicators derived from corporate bond data could be used to predict bond ETFs, commodity futures data could predict commodity ETFs, or, with the emergence of tokenized real-world assets (RWA), such as tokenized real estate portfolios or infrastructure projects, technical data from the underlying assets could be used to forecast token returns. Finally, exploring deep learning models, such as long short-term memory (LSTM) or transformer architectures, may uncover nonlinear and temporal patterns that further improve cross-sectional return rankings.

CRedit authorship contribution statement

Zihan Gong: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation. **Sebastian Müller:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization, Methodology.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT to improve the readability of the paper. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A

Table A.1

Overview of technical indicators.

Volume indicator	Volatility indicator	Trend indicator	Momentum indicator	Other indicator
volume_adi	volatility_bbm	trend_macd	momentum_rsi	others_dr
volume_obv	volatility_bbh	trend_macd_signal	momentum_stoch_rsi	others_dlr
volume_cmf	volatility_bbl	trend_macd_diff	momentum_stoch_rsi_k	others_cr
volume_fi	volatility_bbw	trend_sma_fast	momentum_stoch_rsi_d	others_dr_rel
volume_em	volatility_bbp	trend_sma_slow	momentum_tsi	others_dlr_rel
volume_sma_em	volatility_bbhi	trend_ema_fast	momentum_uo	others_cr_rel
volume_vpt	volatility_bbli	trend_ema_slow	momentum_stoch	
volume_vwap	volatility_kcc	trend_vortex_ind_pos	momentum_stoch_signal	
volume_mfi	volatility_kch	trend_vortex_ind_neg	momentum_wr	
volume_nvi	volatility_kcl	trend_vortex_ind_diff	momentum_ao	
volume_adi_rel	volatility_kcw	trend_trix	momentum_roc	
volume_obv_rel	volatility_kcp	trend_mass_index	momentum_ppo	
volume_cmf_rel	volatility_kchi	trend_dpo	momentum_ppo_signal	
volume_fi_rel	volatility_kcli	trend_kst	momentum_ppo_hist	
volume_em_rel	volatility_dcl	trend_kst_sig	momentum_pvo	
volume_sma_em_rel	volatility_dch	trend_kst_diff	momentum_pvo_signal	
volume_vpt_rel	volatility_dcm	trend_ichimoku_conv	momentum_pvo_hist	
volume_vwap_rel	volatility_dcw	trend_ichimoku_base	momentum_kama	
volume_mfi_rel	volatility_dcp	trend_ichimoku_a	momentum_rsi_rel	
volume_nvi_rel	volatility_atr	trend_ichimoku_b	momentum_stoch_rsi_rel	
	volatility_ui	trend_stc	momentum_stoch_rsi_k_rel	
	volatility_bbm_rel	trend_adx	momentum_stoch_rsi_d_rel	
	volatility_bbh_rel	trend_adx_pos	momentum_tsi_rel	
	volatility_bbl_rel	trend_adx_neg	momentum_uo_rel	
	volatility_bbw_rel	trend_cci	momentum_stoch_rel	
	volatility_bbp_rel	trend_visual_ichimoku_a	momentum_stoch_signal_rel	
	volatility_bbhi_rel	trend_visual_ichimoku_b	momentum_wr_rel	
	volatility_bbli_rel	trend_aroon_up	momentum_ao_rel	
	volatility_kcc_rel	trend_aroon_down	momentum_roc_rel	
	volatility_kch_rel	trend_aroon_ind	momentum_ppo_rel	
	volatility_kcl_rel	trend_psar_up	momentum_ppo_signal_rel	
	volatility_kcw_rel	trend_psar_down	momentum_ppo_hist_rel	
	volatility_kcp_rel	trend_psar_up_indicator	momentum_pvo_rel	
	volatility_kchi_rel	trend_psar_down_indicator	momentum_pvo_signal_rel	
	volatility_kcli_rel	trend_macd_rel	momentum_pvo_hist_rel	
	volatility_dcl_rel	trend_macd_signal_rel	momentum_kama_rel	
	volatility_dch_rel	trend_macd_diff_rel		
	volatility_dcm_rel	trend_sma_fast_rel		
	volatility_dcw_rel	trend_sma_slow_rel		
	volatility_dcp_rel	trend_ema_fast_rel		
	volatility_atr_rel	trend_ema_slow_rel		
	volatility_ui_rel	trend_vortex_ind_pos_rel		
		trend_vortex_ind_neg_rel		
		trend_vortex_ind_diff_rel		
		trend_trix_rel		
		trend_mass_index_rel		
		trend_dpo_rel		
		trend_kst_rel		
		trend_kst_sig_rel		
		trend_kst_diff_rel		
		trend_ichimoku_conv_rel		
		trend_ichimoku_base_rel		
		trend_ichimoku_a_rel		
		trend_ichimoku_b_rel		
		trend_stc_rel		
		trend_adx_rel		
		trend_adx_pos_rel		
		trend_adx_neg_rel		
		trend_cci_rel		
		trend_visual_ichimoku_a_rel		
		trend_visual_ichimoku_b_rel		
		trend_aroon_up_rel		
		trend_aroon_down_rel		
		trend_aroon_ind_rel		
		trend_psar_up_rel		
		trend_psar_down_rel		

(continued on next page)

Table A.1 (continued).

Volume indicator	Volatility indicator	Trend indicator	Momentum indicator	Other indicator
		trend_psar_up_indicator_rel trend_psar_down_indicator_rel		
20(12%)	42(24%)	68(40%)	36(21%)	6(3%)

Table A.2

Hyperparameters for all methods.

	Number of estimators	max depth	max features
RandomForestClassifier	300	1,2,3,4,5,6	3,5,10,20,30,50,'sqrt'
DecisionTree	1	1,2,3,4,5,6	3,5,10,20,30,50,'sqrt'
GBDT/GBRT	300	1,2,3,4,5,6	3,5,10,20,30,50,'sqrt'
XGBoost	300	1,2,3,4,5,6	–

This table provides the hyperparameter search space used for tuning each machine learning model in our analysis. “Number of estimators” refers to the number of trees in ensemble methods. “Max depth” refers to the maximum depth of each tree, and “max features” refers to the number of features considered when looking for the best split, including specific values and the ‘sqrt’ option. XGBoost does not require tuning of max features due to its internal handling of feature subsampling.

Table A.3

Methodology pipeline overview.

Step	Description
1. Data collection	Collect data on individual stocks and ETFs.
2. Data preprocessing	Apply filtering and winsorizing procedures to clean the data.
3. Feature construction	Construct 172 technical indicators from the price and volume data.
4. Train-validation-test split	Split the dataset for model training and evaluation under two model settings.
5. Model training	Train the Random Forest model using the training set.
6. Hyperparameter tuning	Use validation accuracy to tune model hyperparameters.
7. Prediction	Generate outperformance probabilities for ETFs.
8. Evaluation & Portfolio construction	Construct long–short portfolios based on predicted rankings and assess strategy performance.
Robustness checks	Validate results using alternative models (Logit, Decision Tree, GBDT, XGBoost).

This table summarizes the sequential steps involved in the predictive modeling pipeline. It outlines the process from raw data collection and preprocessing, through feature engineering and model training, to portfolio construction and performance evaluation. Robustness checks using alternative classification models are also included to ensure the reliability of results.

Table A.4

Model performance using large stocks as proxy constituents.

Training sample	Long	Short	Long–short
Large stocks	0.75	0.24	0.51** (2.03)

This table reports the performance of the model when the training sample is restricted to larger stocks, which is defined by market value above the cross-sectional median. The models are trained from 1981 until 2004 on stock and evaluated from 2011 until 2022 on ETF. The evaluation period spans 2011–2022 for the stock-trained model. The results are tested on the ETF market. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table A.5
Selected indicators from logistic regression.

Indicators	Long	Short	Long-short
Momentum	0.92%	0.34%	0.58%*** (2.98)
Trend	0.67%	0.41%	0.26% (1.29)
Volatility	0.80%	0.38%	0.42% (1.42)
Volume	0.44%	0.72%	−0.28%** (−2.45)
Others	0.83%	0.08%	0.75%*** (2.78)

This table reports the classification results when we restricted our training data to different categories of stock technical indicators. The models are trained from 1981 until 2004 and evaluated from 2011 until 2022. The evaluation period spans 2011–2022 for the stock-trained model. The results are tested on the ETF market. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table A.6
Robust out-of-sample performance in the extended period (2023–2024).

2023–2024			2011–2024		
Long	Short	Long-short	Long	Short	Long-short
Panel A: ETF raw return					
1.73%	0.42%	1.31% (1.44)	1.12%	0.28%	0.84%*** (3.11)
Panel B: ETF CAPM Alpha					
1.90%	1.01%	0.90% (0.86)	0.29%	−0.40%	0.69%** (2.54)
Panel C: ETF 3 Factor Alpha					
2.09%	0.84%	1.25% (1.00)	0.31%	−0.30%	0.61%** (2.26)
Panel D: ETF 4 Factor Alpha					
0.22%	0.64%	1.55% (1.21)	0.39%	−0.11%	0.50%* (1.75)

This table reports the out-of-sample test on two periods: 2023 to 2024 and 2011 to 2024. The table is divided into four panels. Panel A shows the evaluation of these models when applied to ETF raw return predictions, displaying metrics such as long return, short return, long-short mean return, and mean return. Panels B, C, and D present the evaluation results for risk adjusted ETF raw returns. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% levels, respectively.

Table A.7
ETF quintile portfolios.

Quintiles					Difference
Q1	Q2	Q3	Q4	Q5	Q5-Q1
Panel A: ETF raw return					
0.29%	0.35%	0.53%	0.71%	0.95%	0.66%*** (3.07)
Panel B: ETF CAPM Alpha					
−0.34%	−0.36%	−0.19%	0.00%	0.23%	0.56%*** (2.62)
Panel C: ETF 3-Factor Alpha					
−0.31%	−0.33%	−0.17%	0.00%	0.24%	0.55%** (2.54)
Panel D: ETF 4-Factor Alpha					
−0.20%	−0.31%	−0.15%	0.04%	0.34%	0.53%** (2.27)

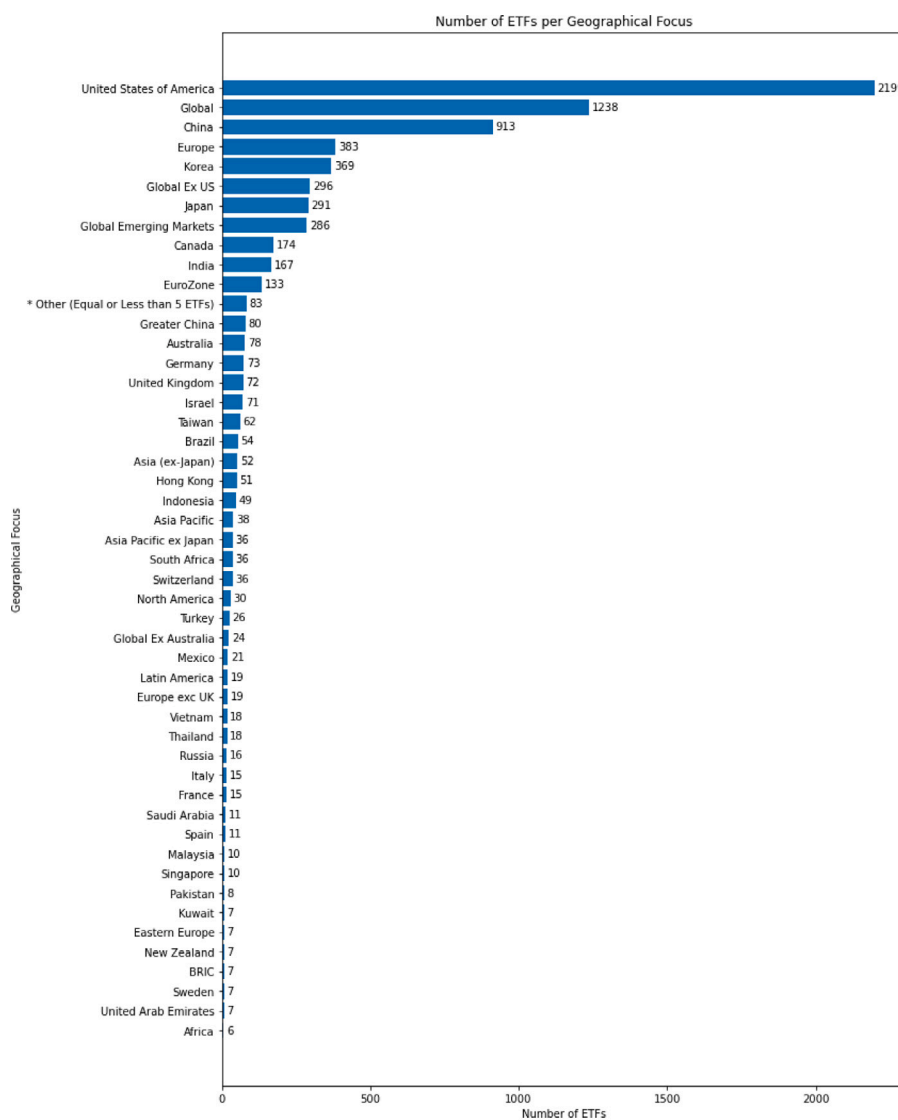
This table reports the raw returns and risk-adjusted alphas of equally weighted ETF portfolios formed by sorting ETFs into quintiles based on predicted outperformance probability. Quintile 1 (Q1) contains ETFs with the lowest predicted returns, while Quintile 5 (Q5) contains those with the highest. Panel A presents raw monthly returns, and Panels B through D report alphas estimated from regressions using the CAPM, Fama–French three-factor model, and Carhart four-factor model, respectively. The final column displays the return or alpha spread between Q5 and Q1 (Q5–Q1), with heteroskedasticity-robust *t*-statistics reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. The sample period spans from January 2011 to December 2022.

Table A.8
ETF vigintile portfolios.

Vigintile										Difference
V1	V2	V3	V4	...	V16	V17	V18	V19	V20	V20–V1
Panel A: ETF raw return										
0.20%	0.32%	0.31%	0.32%	...	0.78%	0.84%	0.91%	0.95%	1.08%	0.88%*** (2.67)
Panel B: ETF CAPM Alpha										
−0.35%	−0.30%	−0.35%	−0.35%	...	0.01%	0.12%	0.21%	0.24%	0.35%	0.70%** (2.14)
Panel C: ETF 3-Factor Alpha										
−0.32%	−0.27%	−0.32%	−0.32%	...	0.01%	0.13%	0.23%	0.25%	0.36%	0.68%** (2.07)
Panel D: ETF 4-Factor Alpha										
−0.15%	−0.18%	−0.21%	−0.25%	...	0.13%	0.20%	0.29%	0.34%	0.51%	0.66%** (1.86)

This table reports the raw returns and risk-adjusted alphas of equally weighted ETF portfolios formed by sorting ETFs into vigintiles (20 groups) based on predicted outperformance probability. Vigintile 1 (V1) contains ETFs with the lowest predicted returns, while Vigintile 20 (V20) contains those with the highest. Panel A presents raw monthly returns, and Panels B through D report alphas estimated from regressions using the CAPM, Fama–French three-factor model, and Carhart four-factor model, respectively. The final column displays the return or alpha spread between V20 and V1 (V20–V1), with heteroskedasticity-robust *t*-statistics reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. The sample period spans from January 2011 to December 2022.

Appendix B



* Other (Equal or Less than 5 ETFs) includes:
 Bangladesh, Europe exc Switzerland, Far East inc Japan, Hungary, Iceland, MENA, Middle East, Peru
 has 1 ETF, Argentina, Australasia, Belgium, Denmark, Egypt, European Misc, Finland, Frontier
 Markets, Portugal has 2 ETFs, GCC, Ireland, Qatar has 3 ETFs, Austria, Far East exc Japan, Greece,
 Nordic, Norway, Philippines, Poland has 4 ETFs, ASEAN, Chile, Colombia, Netherlands has 5 ETFs

Fig. B.1. Number of ETFs by geographical focus.

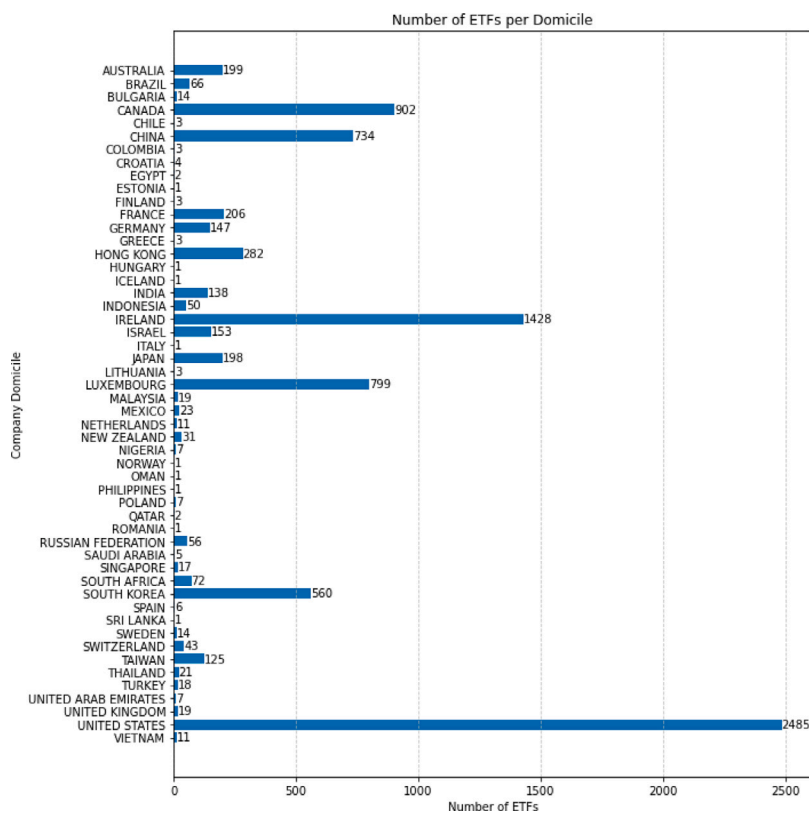


Fig. B.2. Number of ETFs by Domicile.

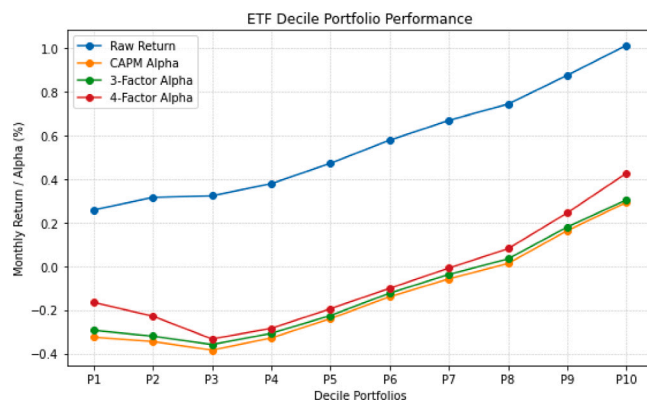


Fig. B.3. ETF decile portfolios (local currency).

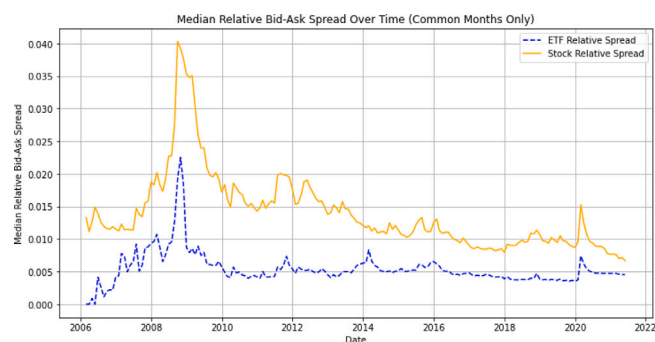


Fig. B.4. Median relative spread over time (Stock vs. ETF).

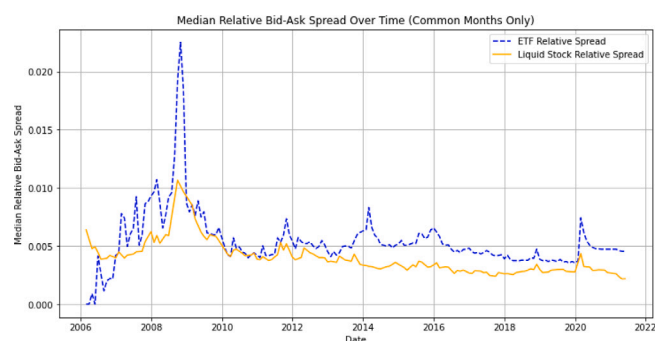


Fig. B.5. Median relative spread over time (Liquid stock vs. ETF).

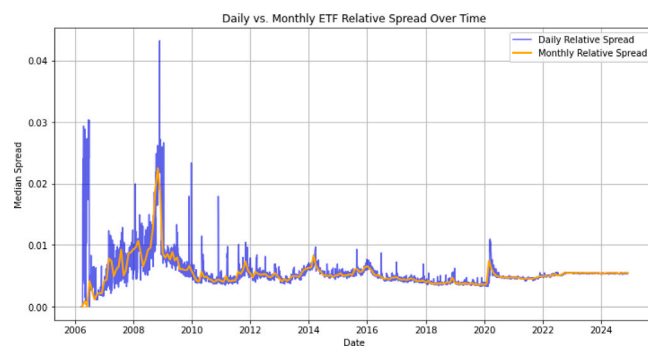


Fig. B.6. Median relative spread over time (Daily vs. Monthly).

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